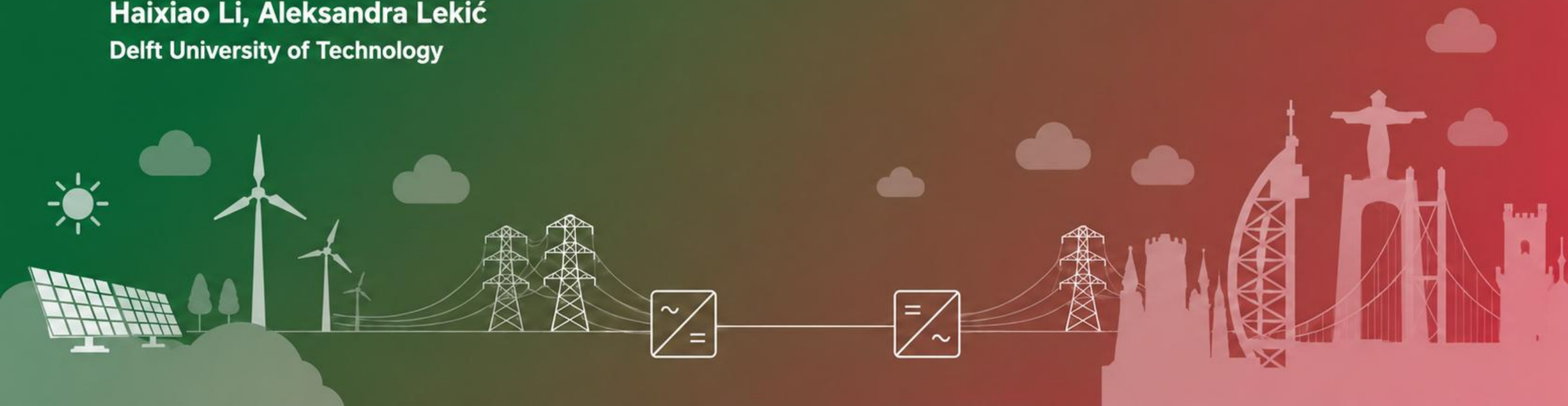
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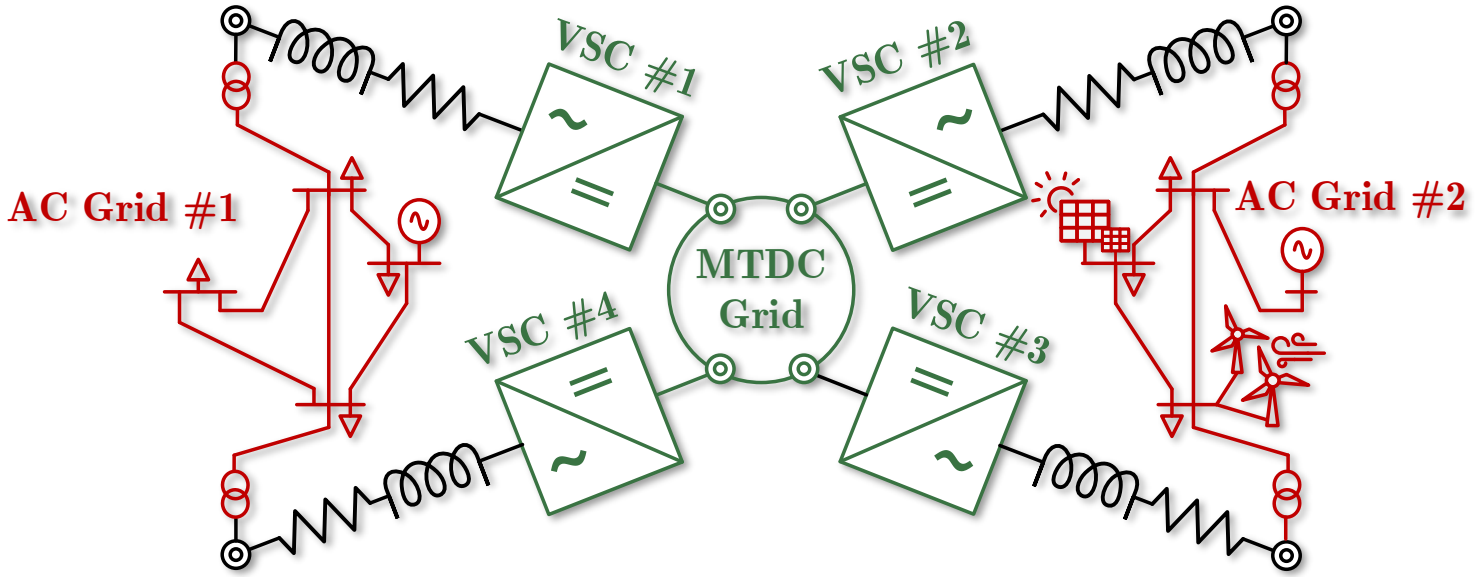
# Leveraging Optimal Information-Power Flow for Transmission Switching in AC/MTDC Grids

**Haixiao Li, Aleksandra Lekić**
**Delft University of Technology**



# 1. Motivation

Leveraging Optimal Information-Power Flow for Transmission Switching in AC/MTDC Grids

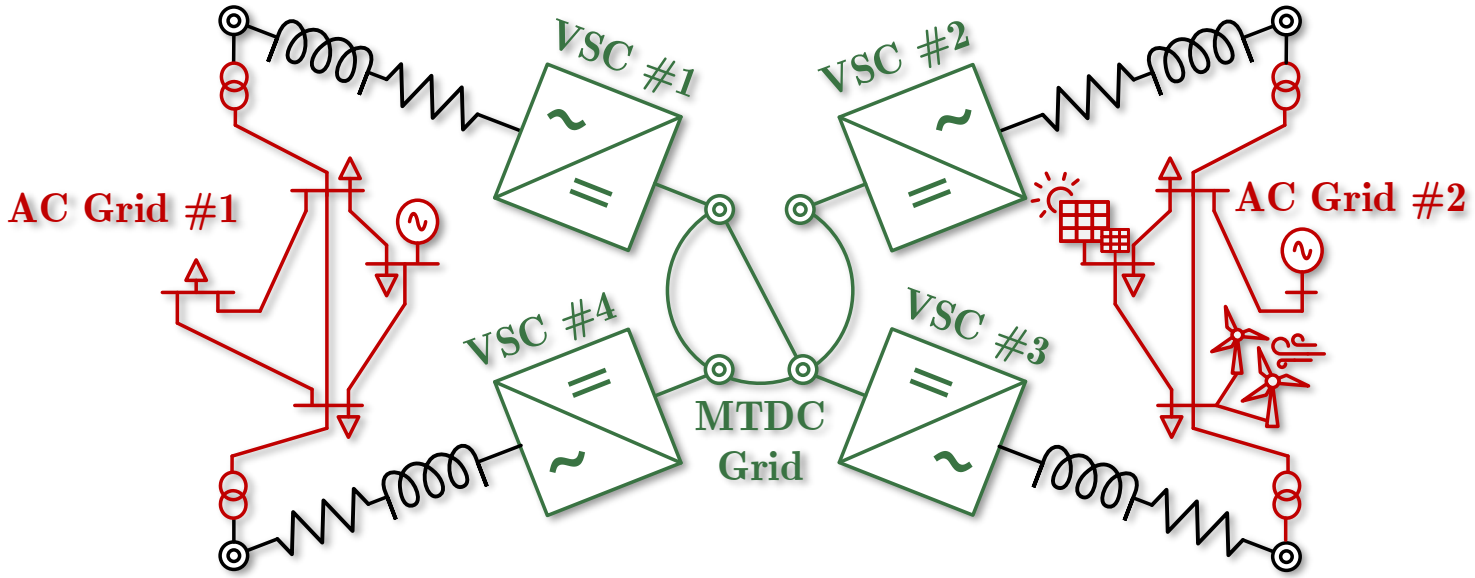



## Operating Considerations

- Achieve the optimal economical operation
- Enhance the flexibility through topology reconfiguration
- Incorporate communication impacts

# 1. Motivation

Leveraging Optimal Information-Power Flow for Transmission Switching in AC/MTDC Grids

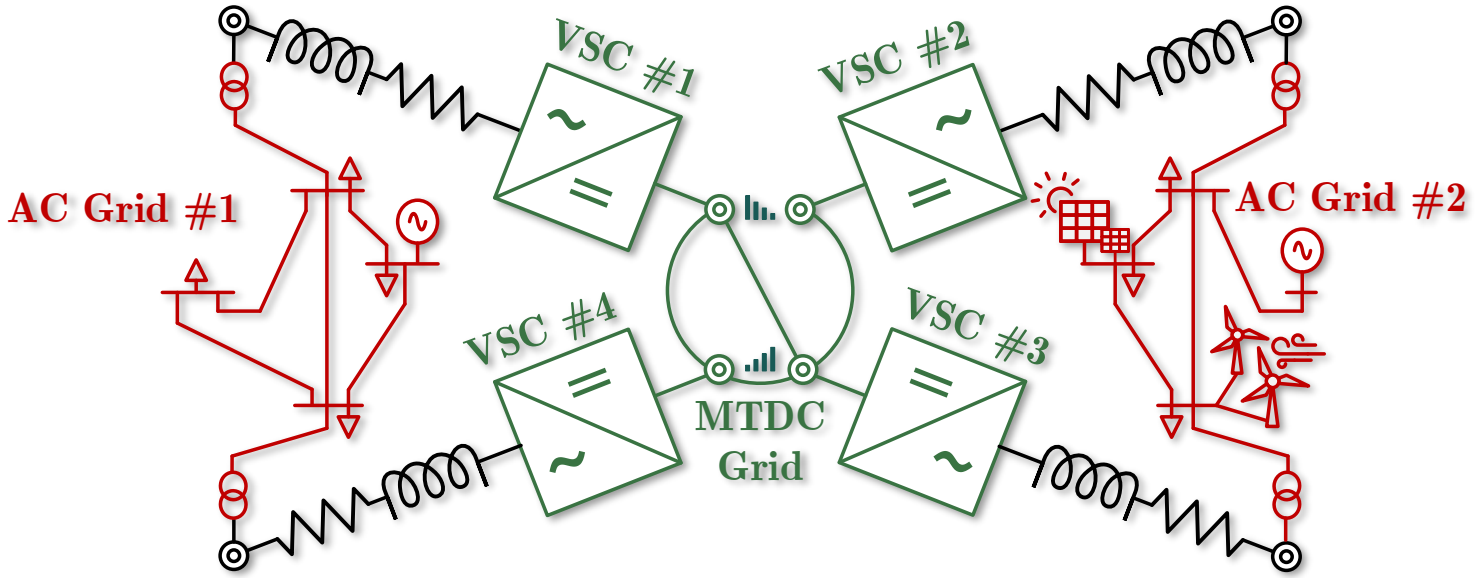



**Operating Considerations**

- Achieve the optimal economical operation
- Enhance the flexibility through DC network reconfiguration
- Incorporate communication impacts

# 1. Motivation

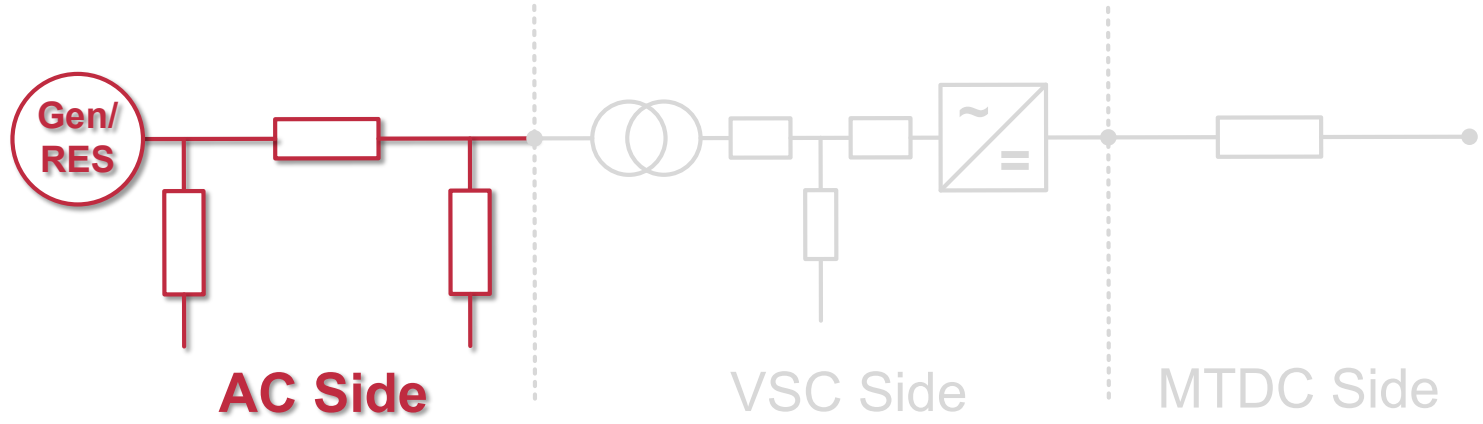
Leveraging Optimal Information-Power Flow for Transmission Switching in AC/MTDC Grids




**Operating Considerations**

- Achieve the optimal economical operation
- Enhance the flexibility through topology reconfiguration
- Incorporate communication impacts

# 2. Model Formulation – Second-order cone (SOCP) relaxed AC/DC power flow constraints




CONSTRAINTS OF AC GRIDS

**AC branches**

$$p_{ij}^{ac} = g_{ij}^{ac}\left(c_{ii}^{ac} - c_{ij}^{ac}\right) + b_{ij}^{ac} s_{ij}^{ac}, \quad q_{ij}^{ac} = -b_{ij}^{ac}\left(c_{ii}^{ac} - c_{ij}^{ac}\right) + g_{ij}^{ac} s_{ij}^{ac}, \qquad \forall (ij) \in \mathcal{E}^{ac}$$ Branch power flow (1)

$$-\bar{s}_{ij}^{ac} \le \cos\left(\frac{k}{N}\pi\right) p_{ij}^{ac} + \sin\left(\frac{k}{N}\pi\right) q_{ij}^{ac} \le \bar{s}_{ij}^{ac}, \quad \forall k \in \{1, 2, \cdots, N\}, \quad \forall (ij) \in \mathcal{E}^{ac},$$ Branch capacity constraint (2)

**AC buses**

$$p_i^{ac} = \sum_{ij} p_{ij}^{ac} + c_{ii}^{ac} G_{i,sh}^{ac}, \quad q_i^{ac} = \sum_{ij} q_{ij}^{ac} - c_{ii}^{ac} B_{i,sh}^{ac}, \quad \forall i \in \mathcal{N}^{ac}, \quad \forall (ij) \in \mathcal{E}^{ac}$$ Bus power injection (3)

$$p_i^{ac} = p_{i,gen}^{ac} - p_{i,load}^{ac} - p_{i,a2v}^{ac}, \quad q_i^{ac} = q_{i,gen}^{ac} - q_{i,load}^{ac} - q_{i,a2v}^{ac}, \quad \forall i \in \mathcal{N}^{ac}$$ Bus power balance (4)

$$\left(\underline{v}_i^{ac}\right)^2 \le c_{ii}^{ac} \le \left(\bar{v}_i^{ac}\right)^2, \quad \forall i \in \mathcal{N}^{ac}$$ Bus voltage bounds (5)

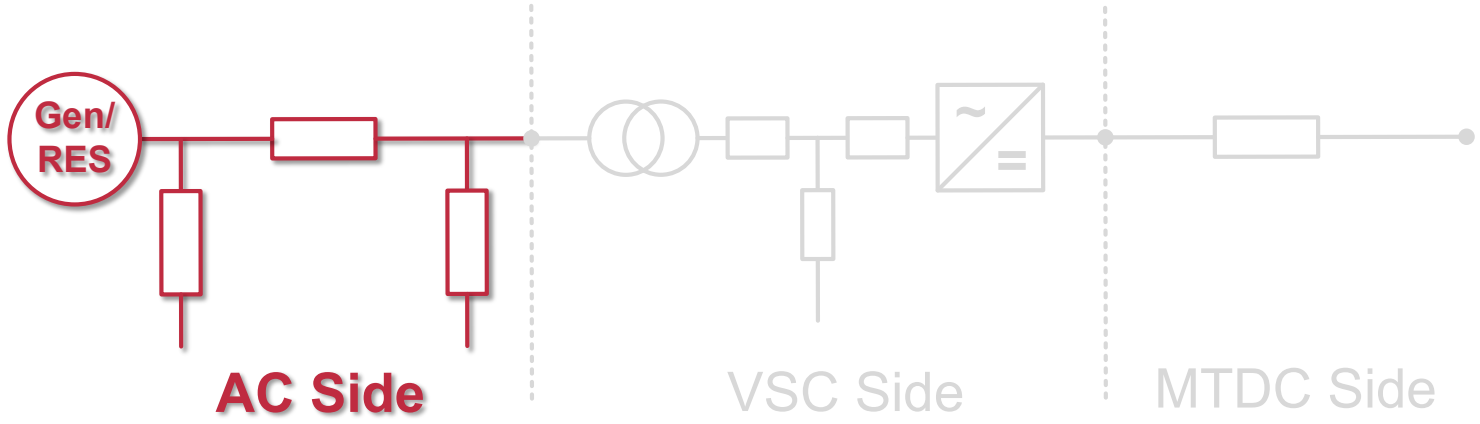
**Generators**

$$\underline{p}_{i,gen}^{ac} \le p_{i,gen}^{ac} \le \bar{p}_{i,gen}^{ac}, \quad \underline{q}_{i,gen}^{ac} \le q_{i,gen}^{ac} \le \bar{q}_{i,gen}^{ac}, \quad \forall i \in \mathcal{N}^{ac}$$ Generator power output limits (6)

**SOCP relaxation**

$$\left(c_{ij}^{ac}\right)^2 + \left(s_{ij}^{ac}\right)^2 \le c_{ii}^{ac} c_{jj}^{ac}, \quad \forall i \in \mathcal{N}^{ac}, \quad \forall (ij) \in \mathcal{E}^{ac}$$ (7)

# 2. Model Formulation – SOCP relaxed AC/DC AC/DC power flow constraints




CONSTRAINTS OF RES PLANTS

**RES power**

$$p_{out}^{res} = p_{r2v}^{res}, \quad q_{out}^{res} = q_{r2v}^{res}, \tag{14}$$

$$0 \le p_{out}^{res} \le \bar{p}_{out}^{res}, \quad -\bar{s}_{out}^{res} \le \cos\left(\frac{k}{N}\pi\right) p_{out}^{res} + \sin\left(\frac{k}{N}\pi\right) q_{out}^{res} \le \bar{s}_{out}^{res}, \quad \forall k \in \{1, 2, \cdots, N\} \tag{15}$$

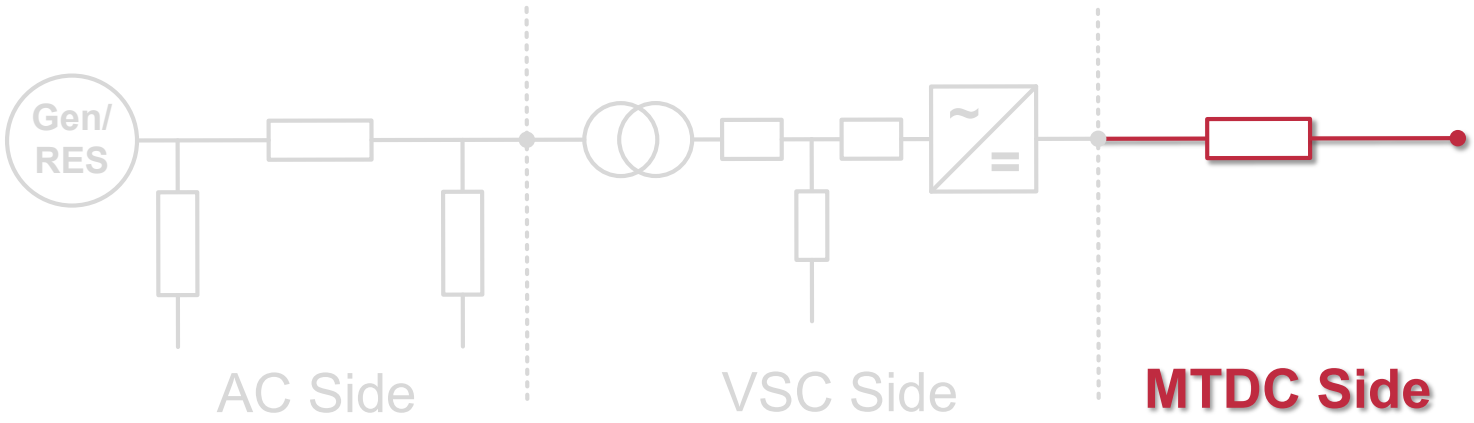
RES power output limits

**RES voltage**

$$\left(\underline{v}_{pcc}^{res}\right)^2 \le u_{pcc}^{res} \le \left(\bar{v}_{pcc}^{res}\right)^2, \quad \forall j \in \mathcal{N}^{res} \tag{16}$$

RES PCC voltage bounds

# 2. Model Formulation – SOCP relaxed AC/DC power flow constraints




CONSTRAINTS OF MTDC GRIDS

**DC branches**

$$p_{jh}^{dc} + p_{hj}^{dc} = r_{jh}^{dc} l_{jh}^{dc}, \quad \forall (jh),(hj) \in \mathcal{E}^{dc} \qquad \text{Branch power flow} \qquad (8)$$

$$u_j^{dc} - u_h^{dc} = r_{jh}^{dc}\left(p_{jh}^{dc} - p_{hj}^{dc}\right), \quad \forall j,h \in \mathcal{N}^{dc}, \quad \forall (jh),(hj) \in \mathcal{E}^{dc} \qquad (9)$$

$$\underline{p}_{jh}^{dc} \le p_{jh}^{dc} \le \bar{p}_{jh}^{dc}, \quad \forall (jh) \in \mathcal{E}^{dc} \qquad \text{Branch capacity limit} \qquad (10)$$

**DC buses**

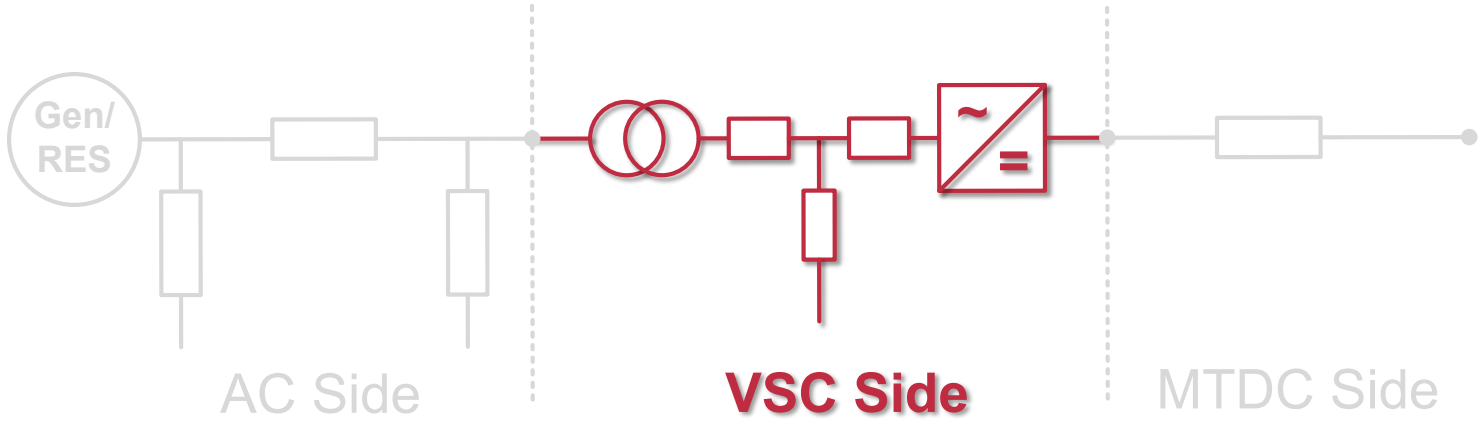
$$p_j^{dc} = \rho^{dc} \sum_{jh} p_{jh}^{dc}, \quad \forall j \in \mathcal{N}^{dc}, \quad \forall (jh) \in \mathcal{E}^{dc} \qquad \text{Bus power injection} \qquad (11)$$

$$\left(\underline{v}_j^{dc}\right)^2 \le u_j^{dc} \le \left(\bar{v}_j^{dc}\right)^2, \quad \forall j \in \mathcal{N}^{dc} \qquad \text{Bus voltage bounds} \qquad (12)$$

**SOCP relaxation**

$$\left(p_{jh}^{dc}\right)^2 \le l_{jh}^{dc} u_j^{dc}, \quad \forall j \in \mathcal{N}^{dc}, \quad \forall (jh) \in \mathcal{E}^{dc} \qquad (13)$$

# 2. Model Formulation – SOCP relaxed AC/DC power flow constraints




CONSTRAINTS OF VSC STATIONS

**VSC Buses**

$$p_s^{vsc} = c_{ss}^{vsc} g_{sc}^{vsc} - c_{sc}^{vsc} g_{sc}^{vsc} + s_{sc}^{vsc} b_{sc}^{vsc}, \quad q_s^{vsc} = -c_{ss}^{vsc} b_{sc}^{vsc} + c_{sc}^{vsc} b_{sc}^{vsc} + s_{sc}^{vsc} g_{sc}^{vsc}, \quad s, c \in \mathcal{N}^{vsc} \tag{17}$$

$$p_c^{vsc} = c_{cc}^{vsc} g_{cs}^{vsc} - c_{cs}^{vsc} g_{cs}^{vsc} + s_{cs}^{vsc} b_{cs}^{vsc}, \quad q_c^{vsc} = -c_{cc}^{vsc} g_{cs}^{vsc} + c_{cs}^{vsc} b_{cs}^{vsc} + s_{cs}^{vsc} g_{cs}^{vsc}, \quad c, s \in \mathcal{N}^{vsc} \tag{18}$$

Bus power injection

$$p_s^{vsc} = \begin{cases} p_{i,a2v}^{ac}, & \text{if } s \text{ is connected to the AC grid} \\ p_{r2v}^{res}, & \text{if } s \text{ is connected to the RES plant} \end{cases}, \quad q_s^{vsc} = \begin{cases} q_{i,a2v}^{ac}, & \text{if } s \text{ is connected to the AC grid} \\ q_{r2v}^{res}, & \text{if } s \text{ is connected to the RES plant} \end{cases}, \quad s \in \mathcal{N}^{vsc}, \quad i \in \mathcal{N}^{ac} \tag{19}$$

Bus power coupling

**VSC Power Loss**

$$p_c^{vsc} + p_{loss}^{vsc} + p_j^{dc} = 0, \quad c \in \mathcal{N}^{vsc}, \quad j \in \mathcal{N}^{dc} \tag{20}$$

Converter power loss

$$p_{loss}^{vsc} = a_{loss}^{vsc} + b_{loss}^{vsc} I_c^{vsc} + c_{loss}^{vsc} l_c^{vsc}, \quad c \in \mathcal{N}^{vsc} \tag{21}$$

$$0 \le I_c^{vsc} \le \bar{I}_c^{vsc}, \quad 0 \le l_c^{vsc} \le (\bar{I}_c^{vsc})^2, \quad c \in \mathcal{N}^{vsc} \tag{22}$$

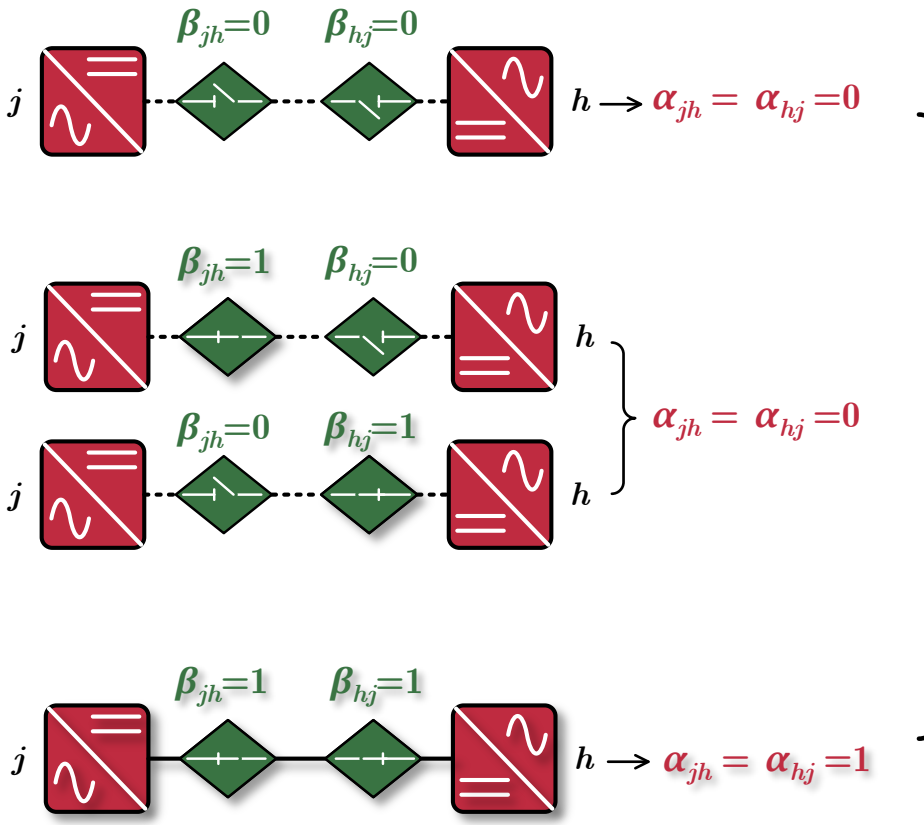
Converter current bounds

**SOCP Relaxation**

$$(I_c^{vsc})^2 \le l_c^{vsc}, \quad (p_c^{vsc})^2 + (q_c^{vsc})^2 \le c_{cc}^{vsc} l_c^{vsc}, \quad c \in \mathcal{N}^{vsc} \tag{23}$$

# 2. Model Formulation – Constraints for DC Transmission Switching

▶ What is the relation between breaker switching and branch switching ('•.•')?




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▶ Relation between breaker switch and branch switch:

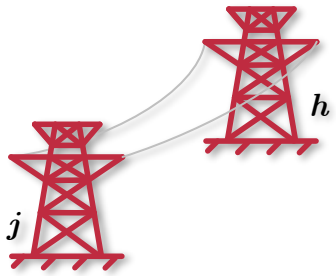
$$\begin{cases} \alpha_{jh} \le \beta_{jh} \\ \alpha_{jh} \le \beta_{hj} \\ \alpha_{jh} \ge \beta_{jh} + \beta_{hj} - 1 \end{cases}$$

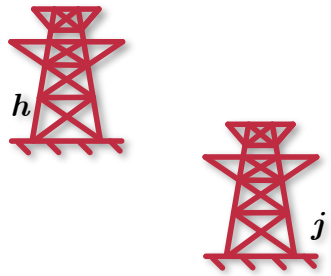
$\alpha_{jh}$ – Represents DC branch status: open (1) / close (0)  $\beta_{jh}$ – Represents breaker status: open (1) / close (0)

# 2. Model Formulation – Constraints for DC Transmission Switching

⏻ Is DC branch **enabled**

☑ ☒




▶ Unified formulation for DC power flow incorporating DC transmission switching

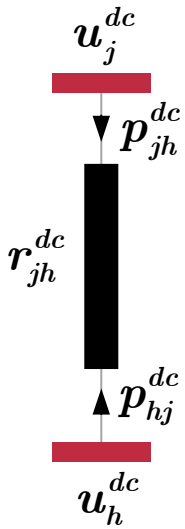
With $\alpha_{jh} = \alpha_{hj} = 1$

$$-\mathrm{M} \le p_{jh} \le \mathrm{M}$$
$$(u_j - b_{jh}) - (u_h - t_{jh}) = r_{jh}(p_{jh} - p_{hj})$$
$$0 \le b_{jh} \le 0$$
$$0 \le t_{jh} \le 0$$
$$(\underline{v}_j)^2 \le u_j \le (\bar{v}_j)^2$$
$$(\underline{v}_h)^2 \le u_h \le (\bar{v}_h)^2$$

$$-\mathrm{M}\alpha_{jh} \le p_{jh} \le \mathrm{M}\alpha_{jh}$$
$$(u_j - b_{jh}) - (u_h - t_{jh}) = r_{jh}(p_{jh} - p_{hj})$$
$$(\underline{v}_j)^2(1-\alpha_{jh}) \le b_{jh} \le (\bar{v}_j)^2(1-\alpha_{jh})$$
$$(\underline{v}_h)^2(1-\alpha_{jh}) \le t_{jh} \le (\bar{v}_h)^2(1-\alpha_{jh})$$
$$(\underline{v}_j)^2 \alpha_{jh} \le u_j - b_{jh} \le (\bar{v}_j)^2 \alpha_{jh}$$
$$(\underline{v}_h)^2 \alpha_{jh} \le u_h - t_{jh} \le (\bar{v}_h)^2 \alpha_{jh}$$

With $\alpha_{jh} = \alpha_{hj} = 0$

$$0 \le p_{jh} \le 0$$
$$(u_j - b_{jh}) - (u_h - t_{jh}) = r_{jh}(p_{jh} - p_{hj})$$
$$(\underline{v}_j)^2 \le b_{jh} \le (\bar{v}_j)^2$$
$$(\underline{v}_h)^2 \le t_{jh} \le (\bar{v}_h)^2$$
$$0 \le u_j - b_{jh} \le 0$$
$$0 \le u_h - t_{jh} \le 0$$

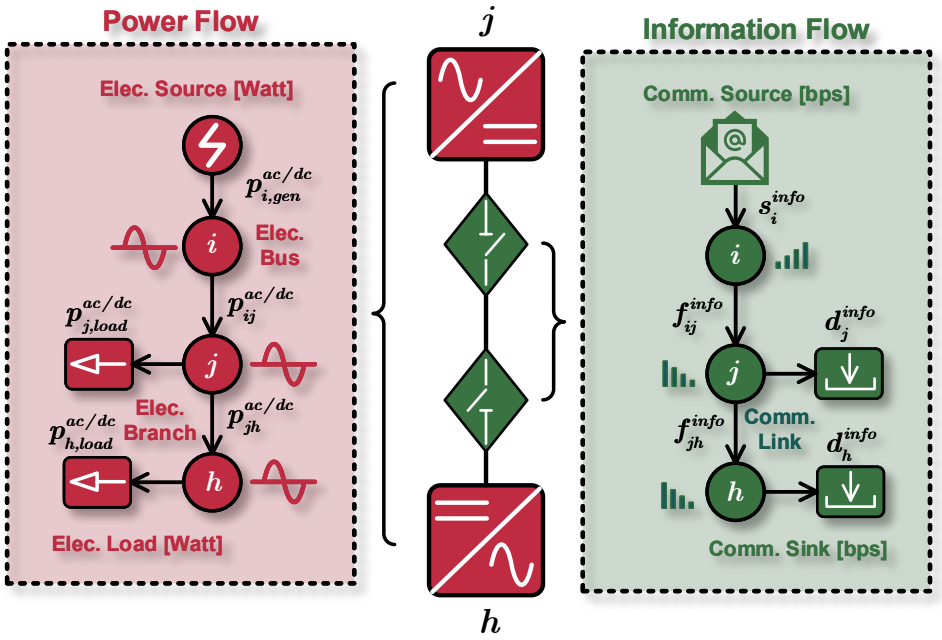



$u_j$ — $p_{jh} = p_{hj} = 0$ — $u_h$

$u_j^{dc}, u_h^{dc}$ – Square of DC bus voltage  $r_{jh}^{dc}$ – Resistance of DC branch  $p_{jh}, p_{hj}$ – DC branch power  $\mathrm{M}$– Large positive number

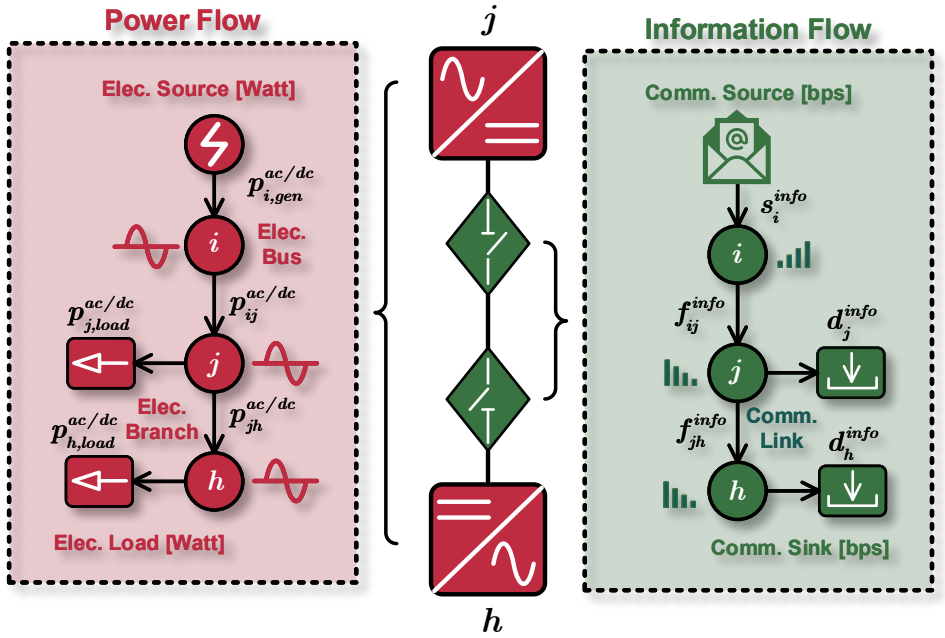
# 2. Model Formulation – Constraints for Information Flow

▶ Information flow is similar to power flow, from a graph theoretic perspective.




Electrical buses ⇄ Communication nodes
Electrical lines ⇄ Communication links
Electrical sources ⇄ Communication sources
Electrical loads ⇄ Communication sinks

# 2. Model Formulation – Constraints for Information Flow




▶ Basic information flow balance formulation

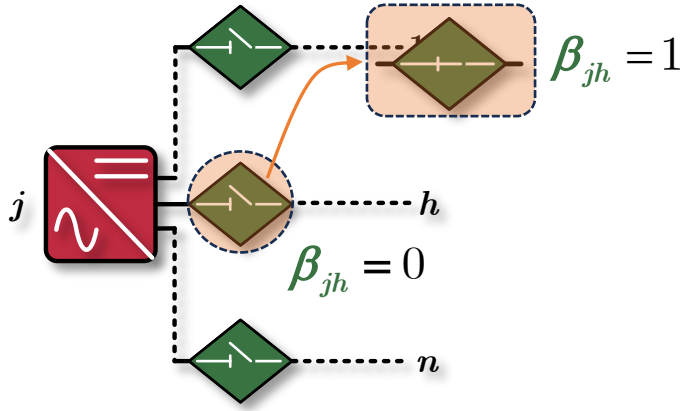
$$\sum_{ij} f^{info}_{ij} = s^{info}_i - d^{info}_i$$

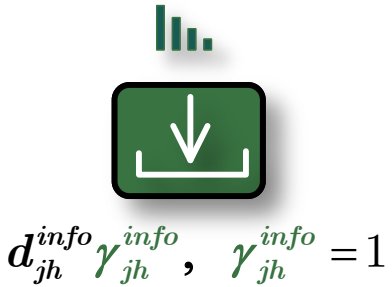
$$-\overline{f}^{info}_{ij} \le f^{info}_{ij} \le \overline{f}^{info}_{ij}$$

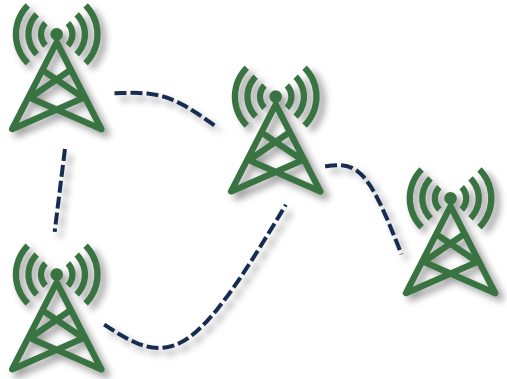
$s^{info}_i$ – Communication source $d^{info}_i$ – Communication sink $f^{info}_{ij}$ – Information flow

# 2. Model Formulation – Constraints for Information Flow

▶ How are power flow and information flow coupled ('•.•')?



1. AC/DC power flow changes relies on changes in breaker status
2. Communication load of control signal that breaker status changed required need to be satisfied
3. Information flow constraint need to be satisfied

$\beta_{jh}$ – Represents breaker status: open (1) / close (0)

$\gamma_{jh}^{info}$ - Represents control signal status: satisfied (1) / unsatisfied (0)

# 2. Model Formulation – Constraints for Information Flow

▶ How are power flow and information flow coupled ('•.•')?

$\beta_{jh}$ – Represents breaker status: open (1) / close (0)

$\gamma_{jh}^{info}$ - Represents control signal status: satisfied (1) / unsatisfied (0)

What is the relation?

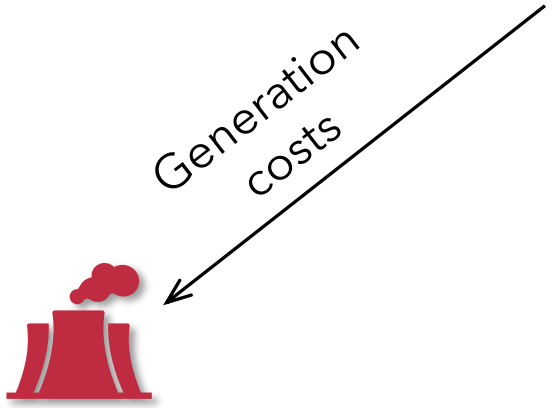
"A breaker status change can only be execute
require
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$$\gamma_{jh}^{info} \geq \beta_{jh} - \hat{\beta}_{jh}, \quad \gamma_{jh}^{info} \geq \hat{\beta}_{jh} - \beta_{jh},$$
$$\gamma_{jh}^{info} \leq \beta_{jh} + \hat{\beta}_{jh}, \quad \gamma_{jh}^{info} \leq 2 - \beta_{jh} - \hat{\beta}_{jh},$$
$$d_j^{info} = \sum_{jh} d_{jh}^{info} \gamma_{jh}^{info}$$

$\hat{\beta}_{jh}$ – Represents pre-decided breaker status: open (1) / close (0)

# 2. Model Formulation – Optimization objectives

▶ Multiple operating costs are involved

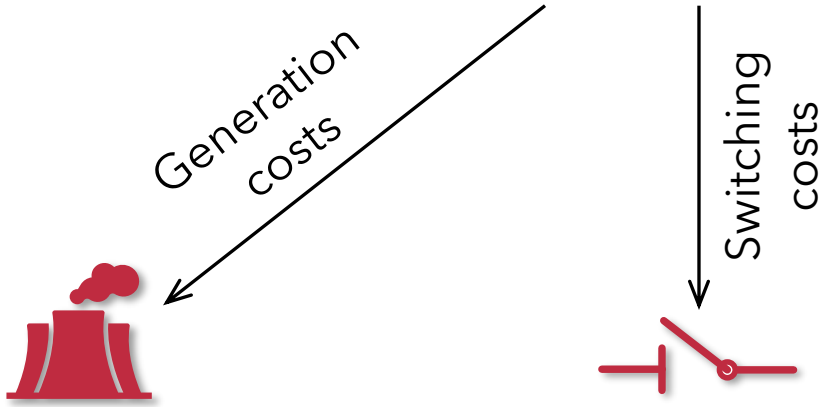



$$\mathbf{Cost}^{st} := \sum_{i} \boldsymbol{a}^{ac}_{i,gen} \left(\boldsymbol{p}^{ac}_{i,gen}\right)^2 + \boldsymbol{b}^{ac}_{i,gen} \boldsymbol{p}^{ac}_{i,gen} + \boldsymbol{c}^{ac}_{i,gen}, \quad \forall \boldsymbol{i} \in \boldsymbol{\mathcal{N}}^{ac}$$

$\boldsymbol{p}^{ac}_{i,gen}$ – Generator power output $\boldsymbol{a}^{ac}_{i,gen}, \boldsymbol{b}^{ac}_{i,gen}, \boldsymbol{c}^{ac}_{i,gen}$ – Quadratic, linear, constant generation cost coefficients $\boldsymbol{\mathcal{N}}^{ac}$ – AC node set

# 2. Model Formulation – Optimization objectives

▶ Multiple operating costs are involved




$$\mathbf{Cost}^{st} := \sum_{i} a^{ac}_{i,gen} \left(p^{ac}_{i,gen}\right)^2 + b^{ac}_{i,gen} p^{ac}_{i,gen} + c^{ac}_{i,gen}, \quad \forall i \in \mathcal{N}^{ac}$$

$$\mathbf{Cost}^{nd} := \sum_{jh} \lambda^{dc}_{jh} \sigma^{dc}_{jh}, \quad \forall (jh) \in \mathcal{E}^{dc}$$

$\lambda^{dc}_{jh}$ – Switching cost for the DC branch $\sigma^{dc}_{jh}$ – Binary indicator for switching cost: ignored (1) / considered (0) $\mathcal{E}^{dc}$ – DC branch set

# 2. Model Formulation – Optimization objectives

▶ Multiple operating costs are involved

Generation costs

Switching costs

Communication costs







$$\mathbf{Cost}^{st} := \sum_{i} a_{i,gen}^{ac} \left(p_{i,gen}^{ac}\right)^2 + b_{i,gen}^{ac} p_{i,gen}^{ac} + c_{i,gen}^{ac}, \quad \forall i \in \mathcal{N}^{ac}$$

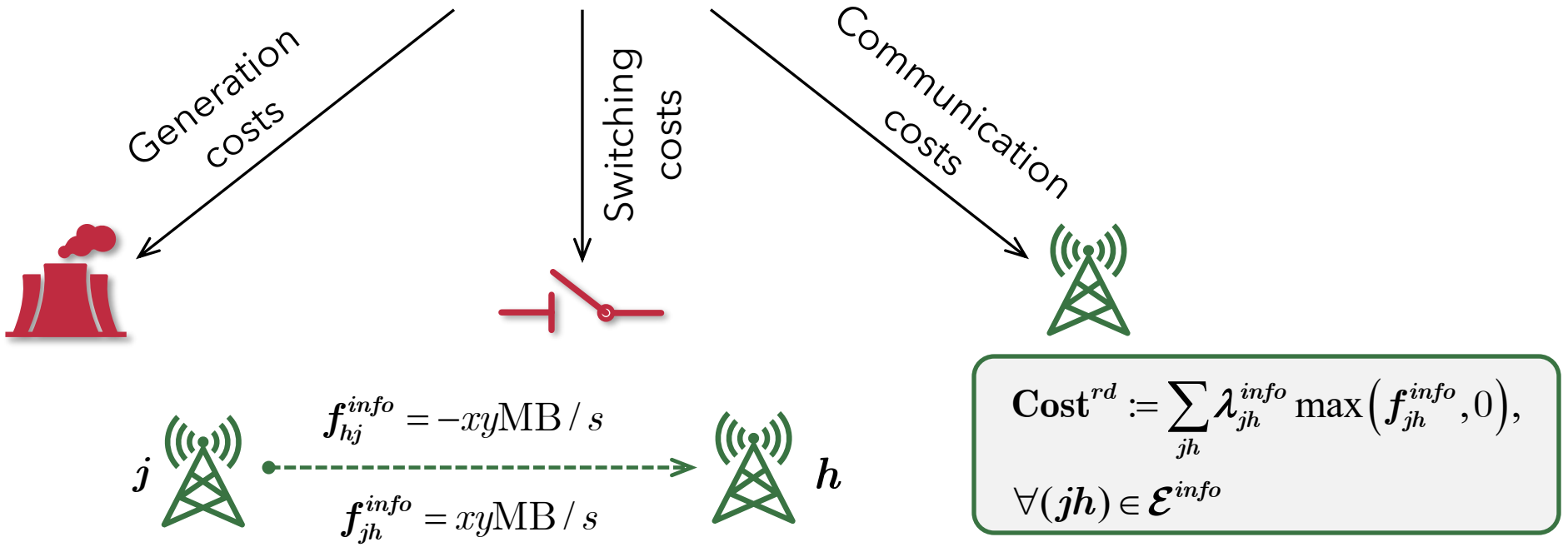
$$\mathbf{Cost}^{nd} := \sum_{jh} \lambda_{jh}^{dc} \sigma_{jh}^{dc}, \quad \forall (jh) \in \mathcal{E}^{dc}$$

$$\mathbf{Cost}^{rd} := \sum_{jh} \lambda_{jh}^{info} \max\left(f_{jh}^{info}, 0\right), \quad \forall (jh) \in \mathcal{E}^{info}$$

$\lambda_{jh}^{info}$ – Communication cost for the communication link $f_{jh}^{info}$ – Communication flow $\mathcal{E}^{info}$ – Communication link set

# 2. Model Formulation – Optimization objectives

▶ Multiple operating costs are involved




$$\mathbf{Cost}^{rd} := \sum_{jh} \lambda_{jh}^{info} \max\left(f_{jh}^{info}, 0\right),$$
$$\forall (jh) \in \mathcal{E}^{info}$$

⚠ Information flow is allowed to take both positive and negative values, but only positive value counted in the objective function

$\lambda_{jh}^{info}$ – Communication cost for the communication link $f_{jh}^{info}$ – Communication flow $\mathcal{E}^{info}$ – Communication link set

# 2. Model Formulation – Optimization objectives

▶ Multiple operating costs are involved

Generation costs

Switching costs

Communication costs







$$\mathbf{Cost}^{rd} := \sum_{jh} \boldsymbol{\lambda}_{jh}^{info} \, \boldsymbol{f}_{jh}^{info,+}, \quad \forall (\boldsymbol{jh}) \in \boldsymbol{\mathcal{E}}^{info}$$

With

$$\boldsymbol{f}_{jh}^{info,+} \geq \boldsymbol{f}_{jh}^{info} \;, \; \boldsymbol{f}_{jh}^{info,+} \geq 0$$

Equivalent

$$\mathbf{Cost}^{rd} := \sum_{jh} \boldsymbol{\lambda}_{jh}^{info} \max\left(\boldsymbol{f}_{jh}^{info}, 0\right),$$

$$\forall (\boldsymbol{jh}) \in \boldsymbol{\mathcal{E}}^{info}$$

$\boldsymbol{\lambda}_{jh}^{info}$ – Communication cost for the communication link $\boldsymbol{f}_{jh}^{info}$ – Communication flow $\boldsymbol{\mathcal{E}}^{info}$ – Communication link set

# 3. Case Studies – Toy case and setting

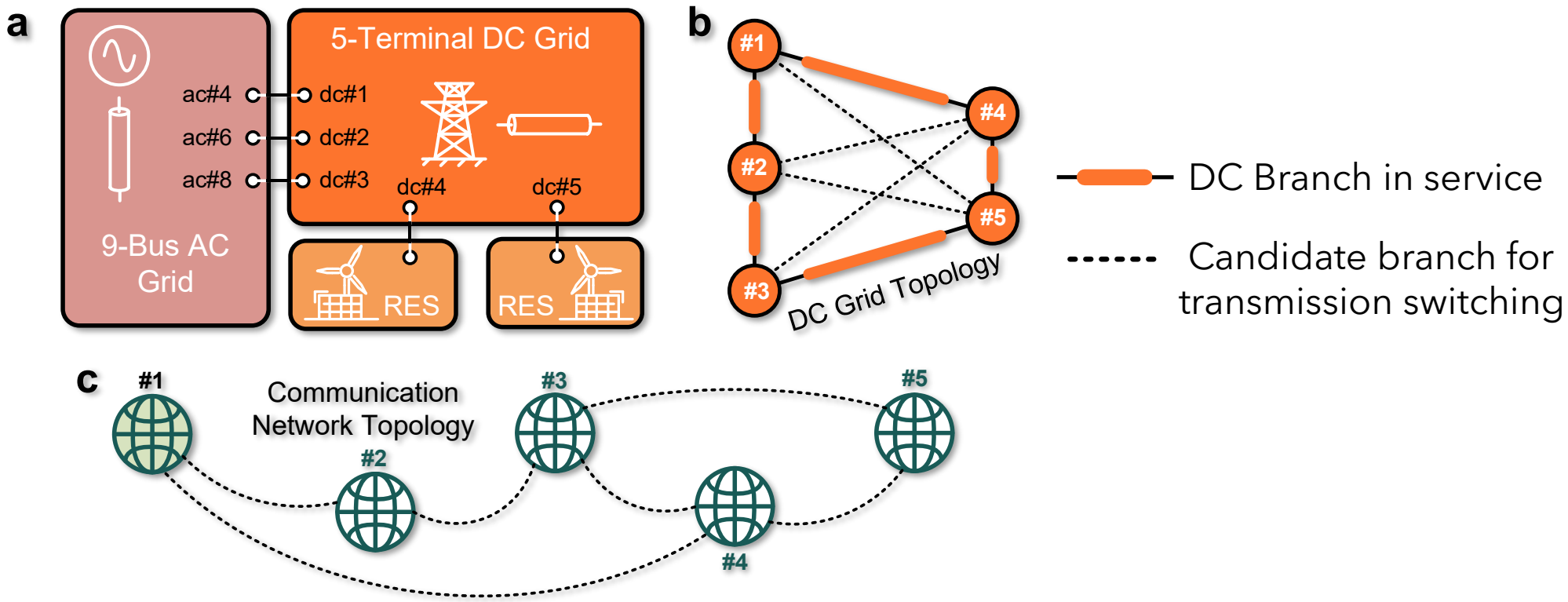



**Fig:** Toy case layout

More details and source code are available @

https://github.com/HLi199305-ee/OIPF-AC-DC-Grids

# 3. Case Studies – Comparison between SOCP-based and NLP-based OPF

▶ Is it **rational** to use the **SOCP relaxed** AC/DC power flow constraints ('•.•')?

| Item | NLP model | SOCP Model |
|---|---|---|
| Generation Cost [ $ ] | 2529.74 | 2518.07 |
| $P_{G1}$[MW] | 43.63 | 43.38 |
| $P_{G2}$[MW] | 77.09 | 76.77 |
| $P_{G3}$[MW] | 54.59 | 54.35 |
| Solution Times [s] | 0.78 | 0.02 |
| Solver | IPOPT | GUROBI |

ⓘ Constraint for DC transmission switching is not included

- Optimal power flow results obtained from the SOCP model are **close** to those of the NLP model („• ᵕ •„)
- SOCP model requires significantly **less computational effort** („• ᵕ •„)

# 3. Case Studies – Comparison between OPF-based and OIPF-based transmission switching

▶ What is the difference between OPF and **optimal Information-power flow** (OIPF) results ('•.•')?

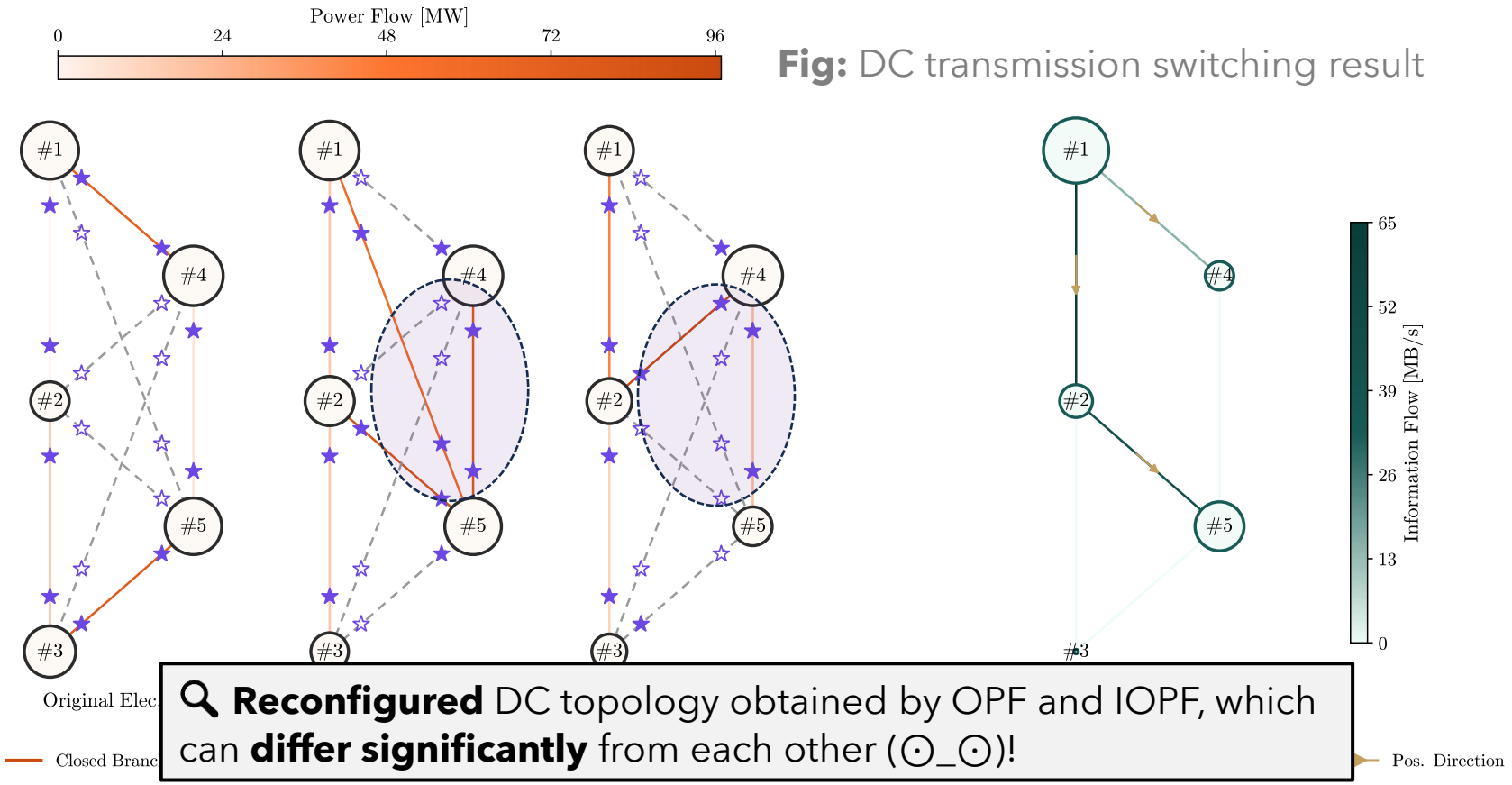



**Fig:** DC transmission switching result

🔍 **Reconfigured** DC topology obtained by OPF and IOPF, which can **differ significantly** from each other (⊙_⊙)!

ⓘ Darker (lighter) branch colors indicate larger (smaller) flow magnitudes, and larger (smaller) node sizes represent higher (lower) injection levels. Both the OPF and OIPF models are formulated as **mixed-integer SOCP** (MISOCP) problems

# 4. Key Takeaways



Based on the considered toy case, the following key takeaways can be drawn

- SOCP relaxation achieves **acceptable accuracy** in power flow and preserves the tractability of the resulting MISOCP-based transmission switching model ʕ̯̍/ˋ˳̫ˊ\
- OIPF considering communication costs can lead to transmission switching decisions that **differ significantly** from those based solely on OPF (⊙_⊙)!
- The communication network modeling **remains preliminary**, and future work will incorporate more realistic communication features, like considering communication delays, to improve the model ( ˘̀_˘́)ↄ

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Mahalo
Gracias
Go raibh maith agat
Danke
Teşekkür ederim
Merci
谢谢
ありがとう
감사합니다
Grazie
Dziękuję
धन्यवाद
Asante
Bedankt
Спасибо
Thank you
Obrigado
Tack
Kiitos
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Ευχαριστώ
Cảm ơn
Salamat

